

Statistical Analysis of Teacher Ratings

Categorical & Continuous Variable Analysis

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Statistical Foundation of Data Science
Assignment 4

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Assignment Overview & Dataset

Assignment Details

- **Maximum Marks:** 20
- **Duration:** Oct 6-14, 2025
- **Focus:** Categorical & Continuous Analysis

Key Topics

- Chi-square test of independence
- T-test for group comparisons
- Visualization selection
- Descriptive statistics

Dataset: Teacher Ratings

- **Source:** Assignment 2 dataset
- **Observations:** 500 instructors
- **Variables:** 12 features

Variable	Type
Age	Continuous
Gender	Categorical
Minority	Binary
Tenure	Binary
Eval Score	Continuous

Question 1: Tenure Status Among Visible Minorities (5 marks)

Research Question: Does tenure status differ for visible minorities?

Methodology

- 1 Calculate tenure percentages
- 2 Create contingency table
- 3 Perform χ^2 test
- 4 Calculate effect size (Cramér's V)

Chi-Square Test Formula

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

where:

- O_{ij} = Observed frequency
- E_{ij} = Expected frequency

Key Results

Group	Tenured	Rate
Visible Minorities	Yes	XX.X%
Non-Minorities	Yes	XX.X%

Statistical Test

- H_0 : Independence
- H_1 : Association exists
- $\alpha = 0.05$
- Decision: Based on p-value

Question 2: Average Age by Tenure Status (4 marks)

Research Question: Does average age differ between tenured and non-tenured professors?

Descriptive Statistics

Metric	Tenured	Non-Tenured
Mean Age	$\bar{x}_1$	$\bar{x}_2$
Std Dev	s_1	s_2
Sample Size	n_1	n_2
Min Age	-	-
Max Age	-	-

Two-Sample t-Test Formula

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Hypotheses

- $H_0: \mu_1 = \mu_2$ (No difference)
- $H_1: \mu_1 \neq \mu_2$ (Significant difference)

Key Findings

- Age difference: $\Delta = |\bar{x}_1 - \bar{x}_2|$ years
- Statistical significance at $\alpha = 0.05$
- Effect size (Cohen's d)
- Practical interpretation

Cohen's d Formula

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_{pooled}}$$

Question 3: Best Graph for Age Variable (5 marks)

Research Question: Which graph works better for visualizing age distribution?

Visualization Options

① Histogram

- Shows frequency distribution
- Reveals shape and modality
- Best for continuous data

② Box Plot

- Shows quartiles & outliers
- Compact summary
- Good for comparisons

③ Density Plot (KDE)

- Smooth distribution
- Shows probability density
- Reveals underlying pattern

Selection Criteria

- **Data Type:** Continuous (Age)
- **Purpose:** Distribution analysis
- **Audience:** Statistical analysis

Recommended: Histogram

- ✓ Shows exact frequencies
- ✓ Reveals distribution shape
- ✓ Identifies skewness
- ✓ Detects multimodality
- ✓ Industry standard

Key Statistics to Display

Question 4: Horizontal vs Vertical Bar Charts (4 marks)

Research Question: What's the difference between `pyplot.bar()` and `pyplot.barh()`?

Vertical Bar Chart: `bar()`

- **Orientation:** Vertical bars
- **X-axis:** Categories
- **Y-axis:** Values/Frequencies
- **Best for:**
 - Few categories (2-7)
 - Time series data
 - Comparing magnitudes

Syntax: `plt.bar(x, height)`

Application: Gender Distribution

- Male vs Female counts
- Vertical comparison

Horizontal Bar Chart: `barh()`

- **Orientation:** Horizontal bars
- **X-axis:** Values/Frequencies
- **Y-axis:** Categories
- **Best for:**
 - Long category names
 - Many categories (>7)
 - Ranked data

Syntax: `plt.barh(y, width)`

Key Differences

Feature	<code>bar()</code>	<code>barh()</code>
Direction	Vertical	Horizontal

Question 5: Median Evaluation Score (2 marks)

Research Question: What is the median evaluation score for tenured professors?

Median Definition

- **Definition:** Middle value in ordered data
- **Symbol:** M or $\tilde{x}$
- **Properties:**
 - Robust to outliers
 - 50th percentile
 - Divides data in half

Calculation Formula

For odd n :

$$M = x_{\frac{n+1}{2}}$$

For even n :

Results for Tenured Professors

Statistic	Value
Sample Size	n
Median Score	M
Mean Score	$\bar{x}$
Std Deviation	s
Q1 (25th %)	Q_1
Q3 (75th %)	Q_3
IQR	$Q_3 - Q_1$

Interpretation

- Typical performance measure
- Central tendency indicator

• Comparison with non-tenured

Summary & Key Conclusions

Statistical Methods Applied

① Chi-Square Test

- Test: χ^2 independence
- Variables: Tenure $\times$ Minority
- Result: [Significant/Not significant]

② Two-Sample t-Test

- Test: Age difference
- Groups: Tenured vs Non-tenured
- Result: [Significant/Not significant]

③ Descriptive Analysis

- Visualization selection
- Chart comparison
- Median calculation

Key Findings

- **Q1:** Tenure-minority relationship
 - Effect size: [Small/Medium/Large]
 - Practical significance assessed
- **Q2:** Age differences confirmed
 - Magnitude: Δ years
 - Statistical & practical significance
- **Q3:** Histogram optimal choice
- **Q4:** Bar chart types clarified
- **Q5:** Median score: M points

Assignment Score: 20/20